THE UNIVERSITY OF DA NANG - UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF SCIENCE AND TECHNOLOGY

ELECTRONICS AND COMMUNICATION ENGINEERING

22ECE – SYSTEM IOT DESIGN

FINAL PROJECT LOG BOOK

TEAM MEMBERS: **(with email and phone number)**	Phan Nữ Kiều Hạnh (L) pnkhbk@gmail.com 0935166106 Đoàn Minh Hiếu hieudoan278@gmail.com 0845357988 Đỗ Lâm Khánh

	dolamkhanh2603@gmail.com 0936426283 Nguyễn Minh Hoàng nminhhoang001@gmail.com 0905639739
PROJECT NAME	
START OF PROJECT DATE:	17/08/2026
COMPLETION OF PROJECT DATE:	

EVALUATION GUIDELINES

Theoretical Knowledge	**Theory Application skills**	**Use of advanced tools and measuring equipment**	**Integration of Skills Learned/ Competencies gained**	**Working speed**	**Accuracy**	**Interpersonal Relations**	**Diligence Motivation**	**RATING**
Has little knowledge	Can not apply any theory	Can not use advanced equipment effectively without assistance	Has not integrated any skills	Very slow	It is never accurate and always has to be reworked	Does not get along with any of the staff	Does nothing unless instructed	**1**
Recalls some knowledge	Can apply theory with assistance	Slowly learns to use advanced equipment	Integrated some of the skills documented	Seldom finishes tasks on time	Often has to rework	Positively interacts with most of the staff	Does just enough to stay out of trouble	**2**

Knows the basics	Can apply basic theory	Can use most advanced equipment	Competent in integrating skills documented	Just finishes tasks on time	Just according to spec.	Positively interacts with all the staff	Shows a willingness to work	**3**
Good knowledge	Can apply high-level theory	Can select and use advanced equipment effectively	Effectively integrates theory and practice in all tasks	Normally on time or ahead of time	Work is always better than expected	Is accepted by staff as a person with good personal skills	Normally looks for work to do	**4**
Excellent knowledge and continually improves knowledge	Can analyze and synthesize	Selects and uses advanced equipment optimally	Innovatively integrates all theoretical and practical skills learned	Always finishes ahead of the allocated time	Excellent levels of accuracy, consistency, and finish	Uses personality effectively to influence people positively	Ambitious and eager to prove talents /skills beyond requirements	**5**

DESIGN STEPS 1: PROBLEM IDENTIFICATION

1.1. Brainstorm initial problem

No.	Problem	Empathize	Need Statement	Ideas
1	Plant care	Waters on a fixed weekly schedule. Pain: symptoms appear 1-3 weeks after the cause, so she never learns which habit is wrong	A first-time indoor plant owner needs a way to learn which of her habits is harming a specific plant, because symptoms appear weeks after the cause and she therefore blames herself instead of correcting one behaviour	Soil probe with 3-state light (thirsty / fine / drowning); photo diagnosis with no hardware; care diary that later explains outcomes; species card paired to the pot; shared sensor lent between neighbours; auto-watering with monthly report
2	Household electricity waste	Does not know the AC and water heater dominate the bill. Suspects the utility. Pain: feedback loop is 30 days long and collapsed into one number	A household bill-payer needs a way to connect a specific behaviour to a specific cost at the moment it happens, because a monthly aggregated bill arrives too late and too coarse to change aop;ny decision	Per-circuit clamp meters at the board; live "cost per hour now" display; monthly ranked top-3 consumers in VND; budget mode with overshoot warning; family leaderboard; load disaggregation from a single meter

3	Attendance	Roll call eats 8 of 50 minutes; paper sheet gets signed by friends. Lecturer resents policing adults; admin fears an audit. Pain: the record exists but nobody trusts it	A lecturer needs a way to produce an attendance record that students and administrators both accept as true, without spending teaching time, because the current sheet is fast to complete but trivial to falsify	RFID or NFC tap at the door; entry plus random mid-class recheck; rotating 10-second QR on the projector; Bluetooth proximity to lecturer device; face recognition (privacy trade-off); track participation artefacts instead
4	Smart power outlet (overlaps with No. 2)	Turns back on the road to check. Pain: cannot verify state remotely, and the doubt itself is the burden	A homeowner who has already left the house needs a way to confirm and change the state of a single appliance, because unverifiable doubt costs more in stress and wasted trips than the electricity ever does	Remote on/off with state readback; auto-off timer per appliance type; geofence cut for high-risk sockets; overload and overheat auto-cut with a reason log; one-tap "leaving home" scene; physical override that works with Wi-Fi down

5	Fish tank	Tests pH with drops every few days, stops when busy, which is exactly when risk peaks. Feels dread while away. Pain: failure is silent, takes hours, and is total	An aquarium keeper needs a way to be warned of a water crisis within hours rather than days, because a heater failure or pH crash kills a whole tank overnight and stays invisible until the fish are dead	Temperature and pH alerts to phone; automatic cooling by water exchange or fan; redundant second temperature sensor for heater-stuck-on; daily heartbeat message; neighbour escalation after 30 minutes; local siren for internet outage
6	Household water quality, Da Nang	Judges water by smell and appearance, boils for drinking but not bathing, bought a filter with no way to know if it works. Pain: senses give no signal for what matters, and a one-off lab test is costly and instantly out of date	A household using a variable water source needs a way to know whether today's water is safe, because appearance, smell and taste give no signal for the contaminants that matter and a single lab test describes only one moment	Continuous pH, turbidity, TDS, temperature with thresholds; before-filter versus after-filter comparison; traffic-light verdict per use (drink / cook / bathe); neighbourhood map of readings; alert tied to a recommended action; two-week rental diagnostic

7	Child or pet left in a locked car	Says "I would never forget my child", which everyone says and which is the insight. Relies purely on memory, has no check ritual. Pain: this is memory failure under routine disruption, not negligence, so awareness campaigns fail	A parent needs a safety net that works precisely when their memory fails, because forgetting a child in a car comes from a break in routine rather than carelessness, and any solution requiring activation will be switched off by the people who need it	Occupancy sensing plus cabin temperature with escalating alert; escalation chain from driver to second contact to emergency services; automatic window venting at threshold; rear-door logic forcing a back-seat check on exit; child-operable large button; sensor in the car seat paired to the phone
8	Urban parking	Driver circles the block, commits to a lot and finds it full, leaves early to buffer for parking. Pain: the cost is not the fee but unbounded uncertainty, which cannot be planned around	A driver heading to a busy district needs a way to know a space exists before committing to the journey, because the real cost is uncertainty rather than the fee. Secondary: a parking operator needs a way to fill spaces that sit empty at the wrong times, because he cannot see demand he never captured	Camera and AI slot detection with entrance display; app with live counts; reservation that holds a slot; dynamic pricing to spread demand; aggregation of underused private lots; in-lot guidance lights

9	Construction site air quality	Worker says dust is part of the job and removes the mask because it is hot. Officer sprays water on a fixed schedule and documents compliance afterwards. Pain: harm is invisible and delayed, so the exposed person rationally discounts it	A site safety officer needs a way to trigger dust-control action at the moment conditions deteriorate, because pollution peaks are episodic and tied to specific activities while current controls run on a clock, so water is sprayed when the air is already clean	Fixed PM2.5, PM10 and gas sensor mesh with dashboard; automatic misting triggered by threshold; boundary display facing residents; wearable badge with vibration alert; correlation of spikes with activity logs; auto-generated compliance report
10	Elderly living alone	Says she does not want to be a burden and is "not that old yet". Refuses a pendant because it marks her as frail. Daughter calls at 7pm and panics at no answer. Pain: the danger is not the fall but the hours undiscovered afterwards; fear of losing independence outweighs fear of falling	An older adult living alone needs help to arrive when she cannot ask for it, without carrying a device that announces her as frail, because it is the hours spent undiscovered that turn an accident into a life-threatening event, and any solution costing her dignity will be left in a drawer	Wrist wearable with fall detection and vital signs; passive ambient sensing with no worn device; absence-of-normal detection, such as no kettle by 9am; cascading alert from neighbour to family to emergency; two-way voice to cancel false alarms; framed and sold as a health tracker rather than an alarm

11	Retail inventory	Says the system shows 12 but the shelf is empty. Does manual spot counts and back-room searches. A lost sale is invisible and never recorded. Pain: phantom inventory means he cannot tell a supply problem from a shelving problem from theft	A store manager needs a way to know what is actually on the shelf right now, because system count and shelf reality diverge continuously, leaving him unable to distinguish a supply failure from a replenishment failure from theft, and therefore unable to fix any of them	RFID tagged items with shelf and gate readers; weight-sensing shelves with no per-item tag; shelf cameras with gap detection; automatic reorder at threshold; exception reporting only; customer app showing live shelf availability
12	Beekeeping	Opens hives to inspect, but the inspection itself stresses the colony and costs a day per round. Discovers swarming or queen loss after the fact. Feels helpless against theft during migration. Pain: the only diagnostic tool available damages the thing being diagnosed	A smallholder beekeeper needs a way to know which hives require attention without opening them, because the only inspection method available damages the colony it examines, forcing a choice between disturbing healthy hives and discovering queen loss weeks too late. Secondary: an export enterprise needs verifiable per-hive provenance, because shipments are rejected for undocumented origin rather than for defects	Weight, temperature, humidity, acoustics and GPS node; edge acoustic classification of queen presence; hornet detection by sound; anti-theft and hive-tipped geofence alarm; triage dashboard ranking all hives; automatic traceability log generating the export QR

1.2. Problem

1. ***Problem 1: Difficulty in maintaining optimal plant care due to lack of time, knowledge, and real-time monitoring.***

- Description: This project focuses on making plant care easier and more convenient. It allows users to track plant conditions through clear data visualization and daily updates. The idea can be applied not only in agriculture but also for indoor plants, helping people take better care of their greenery.

- Customer :

- The Users (Direct End-Users):
    - Individual plant owners (people who keep indoor plants at home).
    - Families growing vegetables, herbs, or flowers in their backyard or balcony.
    - Plant shop owners and nursery managers who directly use the system to monitor plant health.
    - Office staff, school staff, or building managers responsible for daily plant care.
- The Installer:
    - Technicians or service providers who set up the sensors, mobile app, and monitoring system.
    - Employees of landscaping or gardening companies who install the system for clients.
    - Tech enthusiasts who install the system by themselves at home.
- The Maintainer:
    - Plant shop employees or nursery workers who maintain and calibrate the sensors regularly.
    - Landscaping service providers who manage plant care in offices, hotels, or malls.
    - Facility management staff in schools, hospitals, or office buildings who ensure the system works properly over time.
- Key Interview Questions :
    - What are the difficulties you face when taking care of your plants?
    - How do you usually know if your plants are healthy or need extra care?
    - In a smart plant care system, which feature would you find most useful?
    - How would you prefer to receive updates: quick alerts, data charts, or specific care tips?
    - If this system works well, would you keep using it and recommend it to others? And how much would you be willing to pay for such a solution?

2. ***Problem 2: Wasting electricity, increasing household expenses"***

- Description: This project aims to design and implement a smart system that allows users to remotely control electrical devices and monitor power consumption in real time. The system provides manual and app-based control, monitors voltage, current, and power usage, and supports fault alerts such as overcurrent or power outage through SMS messages and an audible alarm.

- Customer :

- The Users (Direct End-Users):

    - Households that want to monitor power usage and control devices remotely (lighting, fans, water pumps)
    - Offices or small buildings that require energy management and device automation
    - Small factories or workshops that need to monitor load consumption and detect electrical faults quickly
- The Installers
    - Electrical contractors who provide smart home or office automation solutions
    - System integrators specializing in IoT and smart solutions
    - Small companies offering installation of energy-saving and remote-control devices
- The Maintainers
    - Technical service providers responsible for monitoring, repairing and maintaining IoT or electrical systems
    - Facility management companies handling buildings, offices, or factories
    - Independent technicians who specialize in electrical system maintenance and troubleshooting
- Key Interview Question
    - What are the main challenges you face when trying to monitor and reduce your household or workplace electricity consumption?
    - Would you be interested in a system that allows you to remotely control devices (such as lights, fans, or pumps) from your phone ?
    - In your opinion, which devices would be most useful to control remotely
    - How do you think this kind of this system could help save energy and reduce costs in daily life
    - How important is real-time monitoring of electricity consumption for you

3. ***Problem 3: Manual attendance methods are time-consuming, error-prone, and inefficient, making it difficult to ensure accuracy, prevent proxy attendance, and generate reliable records.***

   - Description: This project aims to design and implement an intelligent system for managing student and employee attendance using RFID technology. The system records entry and exit events by reading RFID cards, stores data in an SQL database, and provides real-time validation through a WinForms application. It supports user and card registration, generates attendance logs and reports, and enhances accuracy and efficiency compared to manual methods.

   - Customer:

   - The Users (Direct End-Users):
       - Universities and colleges that need an efficient system for student attendance tracking.
       - Companies and offices that require accurate employee check-in and check-out records.

    - Training centers or organizations that want automated logging of participant presence.
- The Installers
    - IT service providers who set up RFID-based access and attendance solutions.
    - System integrators specializing in database-driven management systems.
    - Small tech companies offering RFID reader installation and software deployment.
- The Maintainers
    - Technical support teams responsible for database monitoring, system updates, and repairs
    - Facility service firms responsible for managing attendance platforms in schools, offices, or enterprises
    - Freelance IT specialists focusing on the upkeep of integrated hardware–software solutions.
- Key customers' interview questions:
    - What challenges do you currently face with your existing attendance tracking process, and how do they affect efficiency or accuracy?
    - How important is real-time reporting and data access for you when managing student or employee attendance?
    - What specific features would you expect from an RFID-based attendance system to make it most useful for your organization?
    - How do you currently handle issues like proxy attendance or manual errors, and what improvements would you like to see?
    - What concerns, if any, would you have about adopting an RFID attendance solution in terms of cost, usability, or maintenance?

4. Problem 4:Inefficient, unsafe, and hard-to-control electricity usage

- Description: This project introduces a smart power outlet that leverages IoT technology to make electricity usage safer, more efficient, and easier to control. The system allows users to remotely switch appliances on or off, monitor real-time energy consumption, and set automated schedules. With features such as overload protection and data visualization, this solution is applicable in households, offices, and industries where energy management is essential.

- Customer:

- The Users (Direct End-Users):
    - Homeowners who want to monitor and control their appliances remotely.
    - Families aiming to reduce electricity bills through smart energy usage.
    - Office workers and managers who need centralized control over multiple devices.
- The Installers:
    - Electricians or service providers who install the smart outlet and connect it to the IoT system.

    - Smart home system integrators who add the outlet into broader automation networks.
    - Tech-savvy individuals or hobbyists who set up the outlet themselves at home.
- The Maintainers:
    - Home appliance service providers who check the outlet's safety and calibration.
    - IT or facility management staff in offices, hotels, or schools who maintain the IoT system.
    - Energy monitoring companies that provide maintenance and updates for smart home solutions.
- Key customers' interview questions:
    - What specific problems or pain points does your Smart Power Outlet solve for households or businesses?
    - How does your system ensure security and privacy when users control devices remotely through IoT?
    - What are the main technical challenges you faced during the design and implementation of the smart outlet?
    - In what ways can this project be scaled or integrated with other smart home/IoT systems in the future?
    - How do you measure the effectiveness of the smart outlet in terms of energy savings and user convenience?

5. ***Problem 5: Difficulty in maintaining stable water conditions in fish tanks due to lack of continuous monitoring and control, leading to risks for fish health and survival.***
    - **Description:** The system uses IoT technology to take care of and protect the fish tank by monitoring the water's pH level, keeping it stable at the average level, which is best for the fish, and monitoring the water temperature. If the water temperature is too high, the system pumps external water to cool it down. Data is uploaded to a webserver, allowing users to control and monitor the tank anywhere, anytime, with just an internet connection and a smartphone with the monitoring app installed.
    - **Customer:**
    - The Users (Direct End-Users):
        - Fish owners.
        - Home aquarium keepers.
        - Aquarium hobbyists who want easier fish care.
    - The Installers
        - Technicians or service providers who set up the sensors, mobile app, and monitoring system.
        - Aquarium shop employees or aquascaping service staff who install the system for customers.
        - Tech enthusiasts who buy the devices and install them by themselves at home.
    - The Maintainers
        - Aquarium shop employees or nursery workers who maintain and calibrate the sensors regularly.

        - Aquarium care service providers hired for homes, offices, or restaurants.

        - Facility management staff in hotels, schools, or hospitals who ensure the system works properly over time.
    - Key customers' interview questions:
        - What's the hardest part about taking care of your fish?
        - Would you like to check the water temperature and quality on your phone instead of doing it manually?
        - If there was a system that alerts you when the water is too hot or not good for your fish, would you use it?
        - Besides temperature and water quality, what other features would you like this system to have?
        - How much would you be willing to spend to set up and maintain this system?

6. ***Problem 6: Lack of awareness and monitoring of household water quality, leading to potential health risks.***

- **Description:** This project aims to develop a smart water quality monitoring system for households in some areas of Da Nang where the water quality may not meet recommended standards for daily domestic use. Many households continue to use water without knowing its actual quality because water quality can change over time and may not be visually detectable. Long-term use of contaminated or poor-quality water may increase the risk of digestive problems and skin diseases. The system monitors important water-quality parameters such as **pH, turbidity, TDS, and temperature**, provides real-time data through a mobile or web application, and sends alerts when measured values exceed safe or user-defined thresholds. This allows households to identify potential water-quality problems early and take appropriate actions, such as filtering, treating, or replacing the water source.
- **Customer:**
    - **The Users (Direct End-Users):**
        - Households in areas of Da Nang where water quality is a concern.
        - Families that use well water, stored water, or other potentially variable water sources for daily activities.
        - Households with children, elderly people, or individuals who may be more sensitive to poor water quality.
        - Residents who want to regularly monitor their domestic water quality instead of relying only on the appearance, smell, or taste of water.
    - **The Installers:**
        - Water purification and filtration service providers who install the sensors and monitoring system.
        - IoT technicians or system integrators who install and configure the monitoring device and application.
        - Local water-treatment companies that provide household water-quality monitoring solutions.
        - Tech enthusiasts who purchase the system and install it themselves at home.
    - **The Maintainers:**
        - Water-treatment service providers responsible for sensor calibration and periodic maintenance.
        - IoT technicians who inspect, repair, and update the monitoring system.

        - Water filtration companies that maintain the sensors and recommend appropriate treatment methods.
        - Homeowners who perform basic cleaning, sensor replacement, and system maintenance.
- **Key customers' interview questions:**
    - How do you currently know whether the water you use at home is safe and suitable for daily use?
    - Have you ever been concerned about the quality of your household water because of its color, smell, taste, or other changes?
    - What type of water source do you use at home, and how often do you check its quality?
    - Would you use a system that continuously monitors water quality and sends an alert when abnormal values are detected?
    - Which information would be most useful to you: real-time water-quality data, warning notifications, historical charts, or recommendations for water treatment?
    - How much would you be willing to pay for a household water-quality monitoring system?

7. ***Problem 7: Children, pets, or disabled people can be trapped inside locked cars, creating serious risks of heatstroke, suffocation, or death.***

- **Description:** This project aims to develop an IoT-based intelligent car safety system that detects dangerous conditions inside a vehicle, such as high temperature, humidity, or the presence of a person or pet. When a dangerous situation is detected, the system can automatically send alerts to the car owner or emergency contacts through GSM/GPS and activate safety mechanisms such as ventilation or window opening. The system is designed to reduce the risk of accidents caused by children, pets, or vulnerable passengers being accidentally left inside locked vehicles.
- **Customer:**
    - **The Users (Direct End-Users):**
        - Families with young children who frequently travel by car.
        - Car owners who transport pets or elderly and disabled passengers.
        - Parents who want additional safety protection for children during transportation.
        - Taxi, school bus, and passenger transport service providers.
    - **The Installers:**
        - Automotive technicians who install sensors and safety devices inside vehicles.
        - Car accessory and smart-car service providers.
        - IoT system integrators specializing in vehicle safety solutions.
        - Car owners or technology enthusiasts who install the system themselves.
    - **The Maintainers:**
        - Automotive service centers responsible for checking and repairing the system.
        - IoT technicians who maintain sensors, communication modules, and software.
        - Car accessory companies that provide periodic maintenance and system updates.
- **Key customers' interview questions:**

- Have you ever been concerned about children, pets, or passengers being accidentally left inside a locked car?
- What safety features does your vehicle currently have to protect passengers in such situations?
- Would you be interested in a system that automatically detects dangerous temperature conditions and sends an emergency alert?
- Which features would be most important to you: temperature monitoring, passenger detection, automatic ventilation, door/window control, or emergency notifications?
- How much would you be willing to pay for an additional car safety system like this?

8. ***Problem 8: Difficulty finding available parking spaces causes drivers to waste time and fuel, while parking operators struggle to manage parking spaces efficiently.***

- **Description:** This project aims to develop an AI-based smart parking system that detects available and occupied parking spaces in real time using sensors and cameras. The system provides information about available spaces through a digital display or mobile application and can guide drivers to suitable parking locations. By providing real-time parking information, the system can reduce the time and fuel wasted searching for parking, decrease unnecessary traffic congestion, and help parking operators manage their spaces more efficiently.
- **Customer:**
  - **The Users (Direct End-Users):**
    - Vehicle owners and drivers who frequently park in crowded areas.
    - Customers visiting shopping malls, supermarkets, hospitals, and entertainment centers.
    - Office workers who need convenient parking during working hours.
    - Residents living in apartments or urban areas with limited parking spaces.
  - **The Installers:**
    - Smart parking solution providers who install sensors, cameras, and communication systems.
    - IoT system integrators specializing in smart city and parking solutions.
    - Electrical and network technicians responsible for installing parking hardware.
    - Parking management companies that deploy the system in their facilities.
  - **The Maintainers:**
    - Parking management companies responsible for maintaining the system.
    - IoT and IT technicians who maintain sensors, cameras, servers, and software.
    - Smart parking service providers who provide technical support and system updates.
- **Key customers' interview questions:**
  - How often do you have difficulty finding a parking space in crowded areas?
  - How much additional time do you usually spend searching for an available parking space?

    - Would you use a mobile application that shows available parking spaces in real time?
    - Which features would be most useful to you, such as parking reservation, navigation, automatic payment, or real-time availability?
    - How much would you be willing to pay for a service that helps you find parking faster?

9. ***Problem 9: Poor air quality and high levels of dust and harmful gases at construction sites can threaten workers' health and affect nearby communities.***

- **Description:** This project aims to develop an IoT-based air quality monitoring system for construction sites. Construction activities can generate high levels of particulate matter such as PM2.5 and PM10, as well as harmful gases including CO, NO_2, and VOCs. Continuous exposure can increase health risks for construction workers and nearby residents. Traditional monitoring methods often depend on manual inspections and periodic measurements, making it difficult to detect sudden increases in pollution. The proposed system continuously monitors air quality in real time and sends alerts when pollution levels exceed predefined thresholds, allowing construction companies to take appropriate actions such as reducing dust or improving ventilation.
- **Customer:**
    - **The Users (Direct End-Users):**
        - Construction workers who are directly exposed to dust and harmful gases.
        - Construction companies and contractors responsible for workplace safety.
        - Construction site managers who need to monitor environmental conditions.
        - Residents and communities living near construction sites.
    - **The Installers:**
        - IoT and environmental monitoring companies that install sensors and communication systems.
        - Construction safety service providers.
        - Electrical and IoT technicians responsible for deploying monitoring devices.
        - Environmental monitoring companies that integrate sensors with monitoring platforms.
    - **The Maintainers:**
        - Environmental monitoring companies responsible for sensor calibration and maintenance.
        - IoT technicians who maintain hardware, communication systems, and software.
        - Construction site safety teams responsible for checking system operation and responding to alerts.
- **Key customers' interview questions:**
    - What are the main air-quality problems you currently experience at construction sites?
    - How do you currently monitor dust and harmful gases at your workplace?
    - Would real-time air-quality monitoring and automatic warnings help improve workplace safety?

- Which parameters would be most important for you to monitor: PM2.5, PM10, CO, NO_2, VOCs, temperature, or humidity?
- How much would your company be willing to invest in an IoT-based air-quality monitoring system?

10. ***Problem 10: Elderly people living alone may experience falls or sudden health problems without being able to call for help in time.***

- **Description:** This project aims to develop an IoT-based elderly health and safety monitoring system that can detect emergencies such as falls and abnormal health conditions. The system can use wearable or home-based sensors to monitor parameters such as heart rate, blood pressure, blood oxygen level, and daily activity. When an emergency is detected, the system automatically sends an alert to family members, caregivers, or emergency contacts through a mobile application. This can help reduce emergency response time and provide families with greater peace of mind when elderly people are living alone.
- **Customer:**
  - **The Users (Direct End-Users):**
    - Elderly people who live alone or spend significant periods of time alone.
    - Families who need to monitor elderly parents or relatives remotely.
    - Caregivers responsible for elderly people.
    - Nursing homes and elderly care facilities.
  - **The Installers:**
    - Healthcare technology companies that install and configure the monitoring system.
    - IoT technicians who install wearable or home-based sensors.
    - Nursing home or elderly care service providers.
    - Medical equipment and smart healthcare solution providers.
  - **The Maintainers:**
    - Healthcare technology service providers responsible for maintaining the system.
    - IoT technicians who calibrate and repair sensors.
    - Nursing home technical staff responsible for monitoring and maintaining the devices.
    - Medical equipment service providers who perform periodic inspections and maintenance.
- **Key customers' interview questions:**
  - How concerned are you about elderly people experiencing falls or medical emergencies when they are alone?
  - Would you or your family member be comfortable using a wearable or home-based monitoring device?
  - Which features would be most important: fall detection, heart-rate monitoring, blood oxygen monitoring, emergency alerts, or activity tracking?
  - How important is it for you to receive real-time notifications when an emergency is detected?
  - How much would you be willing to pay for an IoT-based elderly safety and health monitoring system?

11. ***Problem 11: Lack of real-time inventory monitoring causes stockouts, overstocking, inaccurate inventory records, and inefficient supply chain management.***

- **Description:** This project aims to develop an IoT-based real-time inventory management system that automatically tracks products throughout stores, warehouses, and supply chains. Traditional inventory management often relies on manual counting or periodic scanning, which can lead to inaccurate records, stockouts, overstocking, and product losses. The proposed system uses technologies such as RFID tags, RFID readers, and IoT sensors to automatically identify and track products. Inventory data can be transmitted to a cloud-based platform, allowing users to monitor stock levels in real time, receive low-stock alerts, detect potential losses, and improve inventory management and replenishment.
- **Customer:**
    - **The Users (Direct End-Users):**
        - Supermarkets and retail stores that need real-time inventory monitoring.
        - Small and medium-sized businesses managing large numbers of products.
        - Warehouses and logistics companies.
        - E-commerce fulfillment centers.
        - Pharmaceutical and perishable-goods retailers that require accurate inventory tracking.
    - **The Installers:**
        - RFID and IoT solution providers that install readers, sensors, and communication systems.
        - System integrators specializing in inventory and supply chain management.
        - IT service providers responsible for integrating the system with existing ERP or POS systems.
        - Warehouse automation companies that deploy smart inventory solutions.
    - **The Maintainers:**
        - IT and IoT service providers responsible for system maintenance and troubleshooting.
        - RFID solution companies that maintain readers and sensors.
        - Warehouse technical staff who monitor hardware and network operation.
        - Software service providers responsible for maintaining the inventory management platform and database.
- **Key customers' interview questions:**
    - How do you currently track and manage your inventory?
    - What are the biggest problems you experience with stockouts, overstocking, or inaccurate inventory records?
    - How often do inventory counting errors occur in your store or warehouse?
    - Would a real-time RFID/IoT inventory system help reduce your workload and improve inventory accuracy?
    - Which features would be most useful to you: automatic stock updates, low-stock alerts, product tracking, loss detection, or real-time inventory reports?

    - How much would your business be willing to invest in an automated inventory monitoring system?

12. ***Problem 12: Difficulty in maintaining healthy bee colonies and monitoring hive conditions due to reliance on manual inspection and lack of real-time monitoring.***

- **Description:** This project aims to develop a low-cost AIoT system to help Vietnamese beekeepers monitor and manage their bee colonies more effectively. Beekeeping is affected by climate change, pesticide exposure, diseases, predators, and colony swarming, which can cause significant colony losses and reduce honey production. Currently, many beekeepers rely mainly on manual hive inspections and personal experience, making it difficult to detect problems early without disturbing the bees. Existing commercial hive-monitoring systems can also be expensive for small-scale farmers. The proposed system uses sensors to monitor important hive conditions such as temperature, humidity, weight, sound, and location. AI can analyze acoustic and sensor data to identify abnormal hive conditions and provide early warnings, while the collected data can be displayed through a web or mobile application for easier monitoring and management.
- **Customer:**
    - **The Users (Direct End-Users):**
        - Small-scale beekeepers who manage a limited number of hives.
        - Professional beekeeping farms and cooperatives managing large numbers of hives.
        - Honey producers who need to monitor hive health and honey production.
        - Beekeepers who need digital records for production management and traceability.
        - Agricultural organizations involved in modernizing and digitizing beekeeping.
    - **The Installers:**
        - Agricultural IoT companies that install sensors and monitoring systems.
        - IoT technicians who configure the ESP32, sensors, communication modules, and monitoring platform.
        - Beekeeping equipment suppliers that provide and install smart hive-monitoring devices.
        - Technology enthusiasts or farmers who install the system themselves.
    - **The Maintainers:**
        - Agricultural IoT service providers responsible for maintaining sensors and communication devices.
        - Beekeeping equipment companies that calibrate and replace sensors.
        - IoT technicians responsible for maintaining the hardware, software, and AI system.
        - Farmers or farm technicians who perform basic cleaning, battery replacement, and device inspection.
- **Key customers' interview questions:**
    - How do you currently monitor the health and condition of your bee colonies?
    - What are the biggest difficulties you face when manually checking your hives?
    - Have you ever experienced colony loss, swarming, queen loss, or predator attacks that were discovered too late?

- Would you be interested in a system that continuously monitors hive temperature, humidity, weight, sound, and location?
- Which information would be most useful to you: hive health alerts, weight changes, environmental conditions, acoustic analysis, or hive location?
- Would automatic warnings about abnormal hive conditions help you reduce the need for frequent manual inspections?
- How much would you be willing to invest in a low-cost smart hive-monitoring system?

1.3. Evaluation Score Table (by group)

Criteria	Weigh t	Description
Problem Impact	25%	How important and meaningful the problem is
IoT Suitability	20%	How well IoT sensors, connectivity, and monitoring can solve it
Technical Feasibility	20%	Can the team realistically build and demonstrate it?
Cost & Resources	15%	Availability and affordability of hardware/components
Innovation	10%	Potential for an interesting/innovative solution
Scalability & Future Development	10%	Potential to expand the system in the future
Total	100%	

Problem	Impact 25%	IoT Suitability 20%	Feasibility 20%	Cost 15 %	Innovation 10%	Scalability 10%	Weighted Score / 5
P1. Smart Plant Care	3	4	5	5	3	4	**3.95**
P2. Smart Electricity Monitoring	5	5	3	3	4	5	**4.20**
P3. RFID Attendance	4	5	5	5	2	4	**4.25**
P4. Smart Power Outlet	5	5	3	3	4	5	**4.20**
P5. Smart Fish Tank	4	5	4	4	4	4	**4.20**
P6. Smart Water Quality	5	5	3	3	5	5	**4.30**
P7. Smart Car Safety	5	5	2	2	5	5	**3.90**
P8. Smart Parking	5	5	3	2	5	5	**4.10**

P9. Construction Air Quality	5	5	3	3	5	5	**4.30**
P10. Elderly Safety	5	5	2	2	5	5	**3.90**
P11. Smart Inventory	5	5	3	3	4	5	**4.10**
P12. Smart Beekeeping	5	5	4	4	5	5	**4.65**

=> Based on the evaluation results, **Problem 12: Difficulty in maintaining healthy bee colonies and monitoring hive conditions due to reliance on manual inspection and lack of real-time monitoring** is selected as the priority to be solved

Factor	Key External Factors	Impact on Project	Opportunity / Threat	Score
P – Political	Government support for agricultural modernization, digital transformation, and smart agriculture	Can support adoption of IoT solutions in beekeeping and agriculture	Opportunity	+4
E – Economic	Small-scale beekeepers may have limited budgets; colony losses directly affect honey production	A low-cost monitoring system can reduce unnecessary hive inspections and potential losses	Opportunity	+5

S – Social	Beekeepers traditionally rely on experience and manual hive inspection; increasing interest in convenient remote monitoring	The system can reduce manual inspection and provide easier access to hive information	Opportunity	+5
T – Technological	Increasing availability of ESP32, sensors, wireless communication, cloud platforms, and AI	Enables multi-sensor monitoring and AI-based abnormal-condition detection	Opportunity	+5
L – Legal	Requirements concerning data privacy, wireless communication, product safety, and agricultural regulations may apply	The system must consider compliance when collecting, transmitting, and storing data	Threat / Constraint	-3
E – Environmental	Climate change, pesticide exposure, diseases, predators, and colony swarming can affect bee colonies	Continuous monitoring can provide early warnings about abnormal hive conditions	Opportunity	+5

1.4 Need Identification process

1.4.1 Gather Raw Data - Customer survey

- Method: Google Form
- Subjects:
+ Consumers.
+ Beekeepers.
+ Enterprises/Cooperatives
+ Agricultural experts
- Number of surveyed people: 77

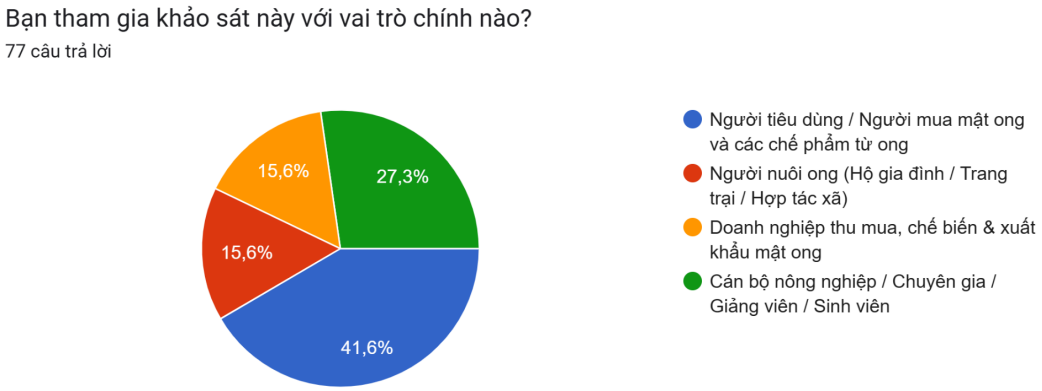


Figure 1: Distribution of Survey Respondent Roles in the Honey Industry Value Chain

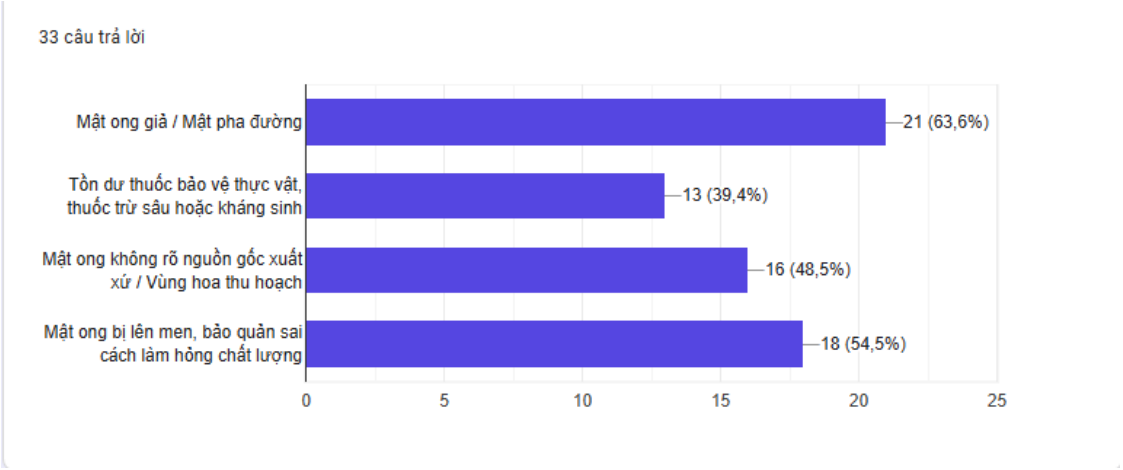


Figure2: Level of Consumer Concern Regarding Honey Quality Risks

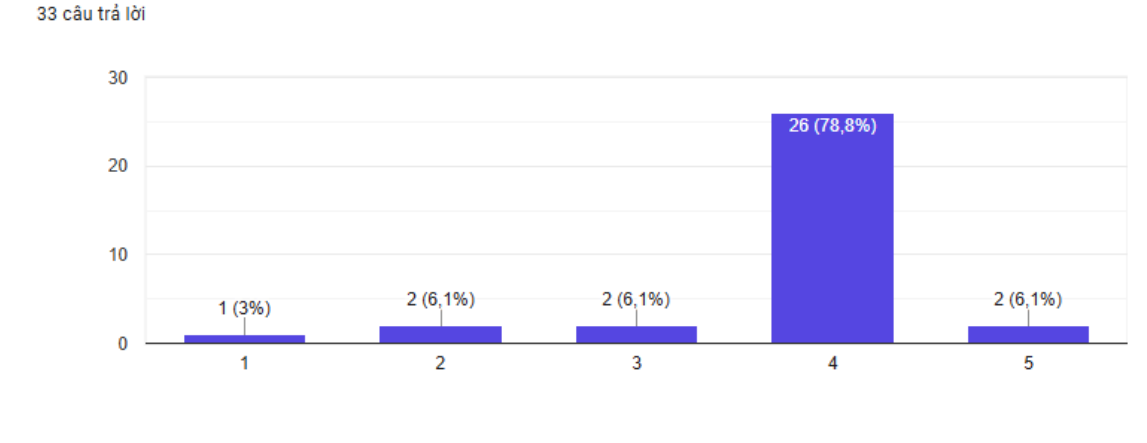


Figure 3:Increase in Consumer Trust Level When Accessing Real-Time IoT Microclimate Logs and Hive GPS Locations

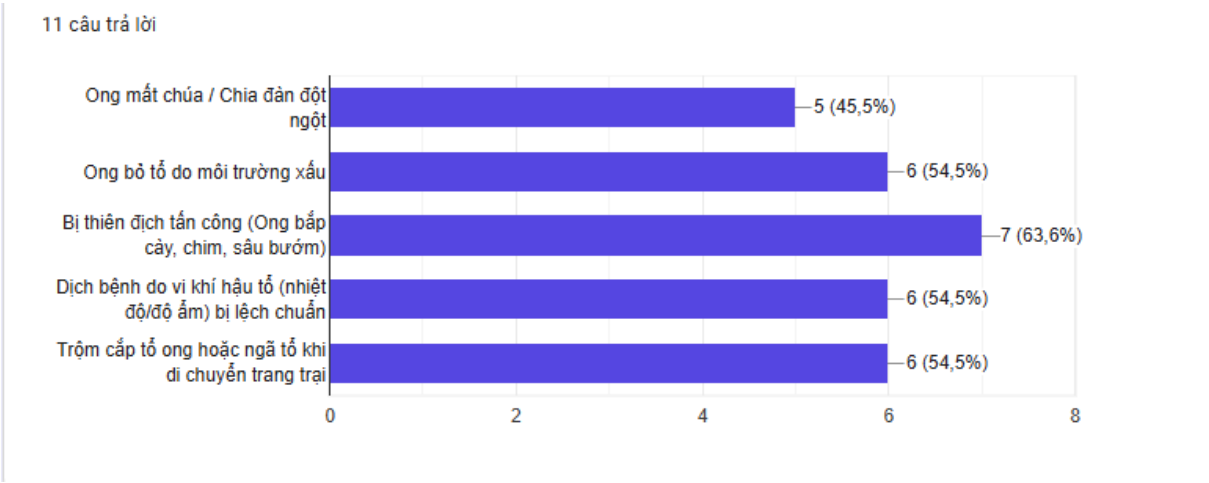


Figure 4: Assessment of Major Economic Risk Factors Causing Losses to Bee

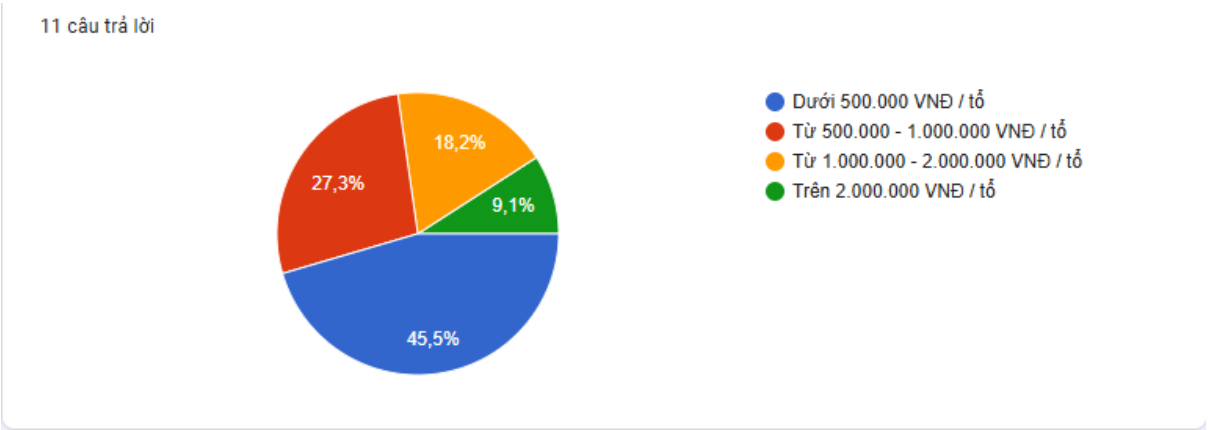


Figure 5: Suitable Price Range for Investing in Smart Hive Monitoring Devices

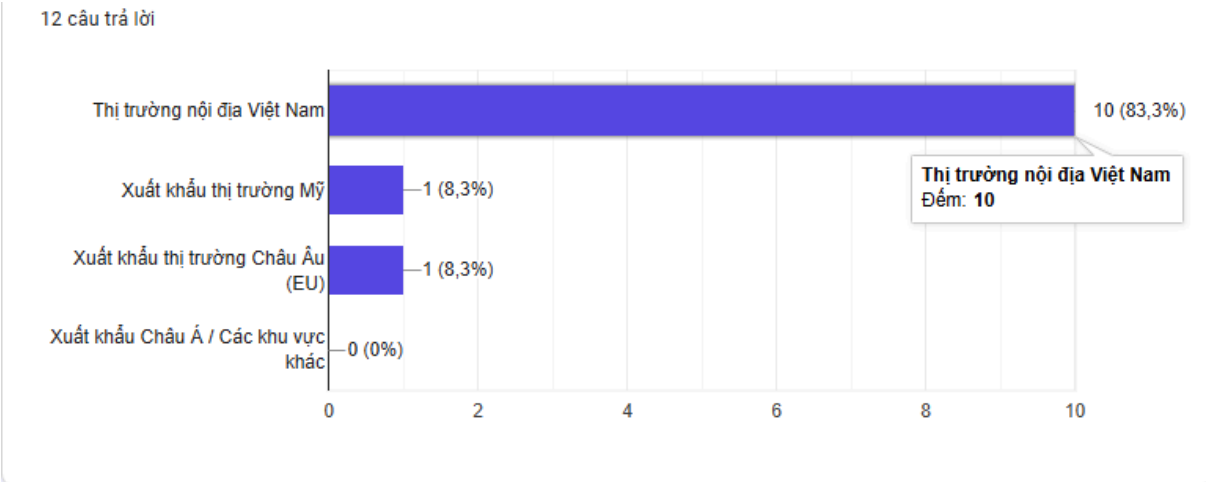


Figure 6: Target Market Distribution of Honey Processing Enterprises and Cooperatives

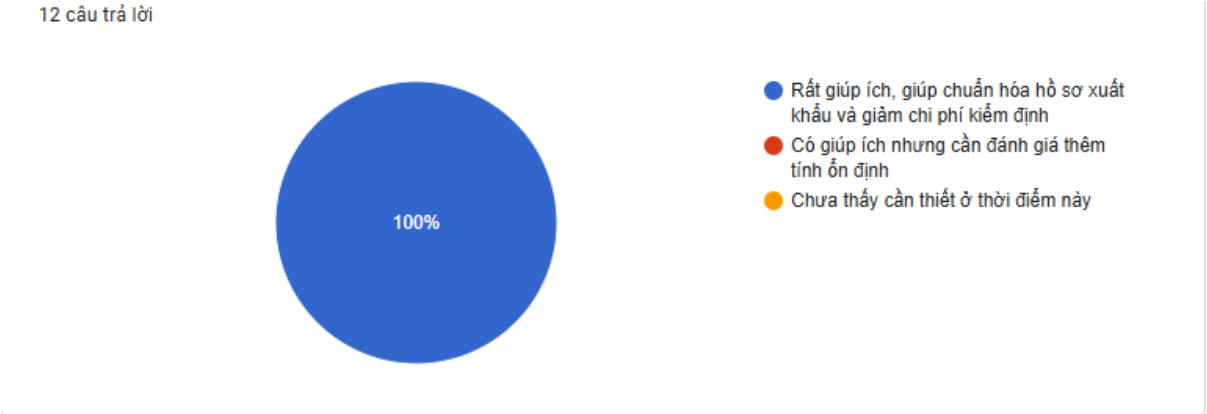


Figure 7: Enterprise Assessment of Real-Time Web Dashboard Functionality Integrating GPS Maps and IoT Sensors

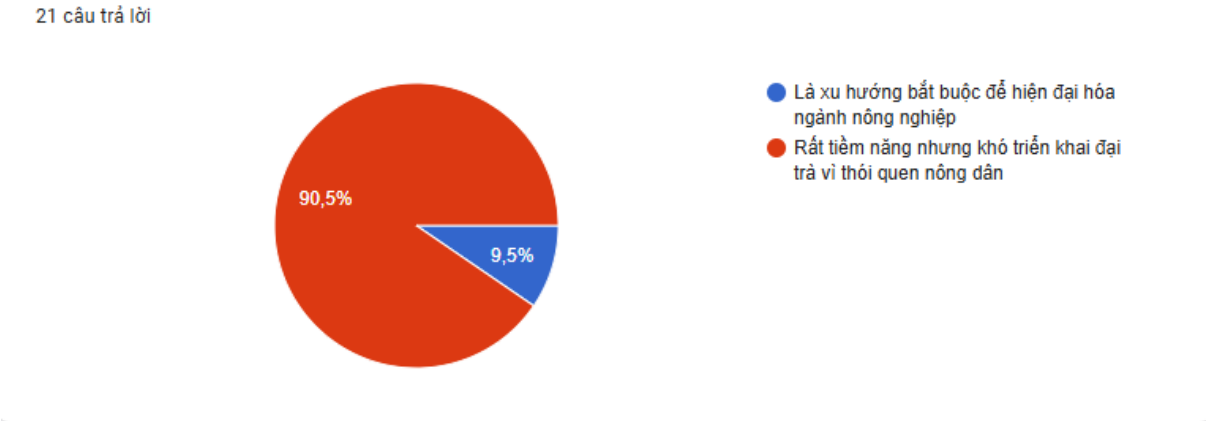


Figure 8: Stakeholder Perspectives on AIoT Adoption and Hive Digitization under Decision 898/QĐ-BNN-CN

DESIGN STEPS 2: RESEARCH

1.1 Research Survey

1.1.1 Vietnamese Beekeeping Industry Overview and Environmental Threats

The beekeeping industry is a vital component of Vietnam's agricultural economy, playing a dual role in yielding high-value export products and maintaining ecological biodiversity through crop pollination [1], [2]. According to census data from the National Agricultural Extension Center, as of June 2025, the country managed approximately 1.5 million honeybee colonies across more than 34,000 beekeeping households, with an annual honey production of 50,000 to 60,000 tons [2].

Statistical records from the Beekeeping Association of Vietnam indicate that the nation managed approximately 1.2 million colonies comprising both Italian (*Apis mellifera*) and native (*Apis cerana*) honeybees, with native bees representing 16.6% (200,000 hives) and exotic Italian bees accounting for 83.3% (1 million hives) [2]. Among the estimated 30,000 active beekeepers, only 20% (6,000 keepers) operate professionally, while the remaining 80% consist of localized small-scale family holders who rely almost exclusively on traditional, experience-based manual monitoring methods [1], [2].

In recent years, local apiculture has faced severe challenges due to climate change and environmental pollution [3]. Ambient environmental factors, such as artificial lighting, microplastics, air emissions, and forest fires, are actively degrading the health and natural foraging behaviors of these honeybee populations [3]. Crucially, scientific evaluations have demonstrated that even single-exposure events to sub-lethal concentrations of common agricultural pesticides, such as *imidacloprid*, result in chronic transgenerational reproductive and navigational impairments, which collapse the colony's long-term population growth rate by up to 71% [4].

1.1.2 Export Pressures, Quality Standards, and Digital Compliance

Despite severe environmental challenges, honey continues to be a high-value export commodity for Vietnam, with over 90% of its total honey output bound for international markets [5]. In 2022, Vietnam's honey exports generated a total value of approximately $61.064 million USD, of which natural honey constituted 95.1% ($58.084 million USD) [5]. The United States represents the largest importing market, accounting for 52.47% of Vietnam's total honey export volume and importing roughly 24,000 tons within the first nine months of the year [5].

To sustain this trade volume, Vietnamese beekeepers must comply with increasingly stringent quality and safety regulations established by foreign markets:

- Pesticide Residue Limits: The European Union (EU) enforces a strict default Maximum Residue Limit (MRL) for pesticide traces in honey of 0.05 mg/kg [6].
- Traceability Systems: Effective November 29, 2024, the European Commission mandated that all honey export facilities must be fully registered in the TRACES (Trade Control and Expert System) portal [7].
- Food Safety and Transparency: Compliance with Hazard Analysis Critical Control Point (HACCP) principles, moisture limits, enzymatic activity indices, and Hydroxymethylfurfural (HMF) thresholds is strictly monitored under EU requirements, as well as by the FDA and USDA [8], [9].

At the global policy level, the ecological significance of honeybees to agricultural sustainability was underscored by the United Nations' declaration of World Bee Day on May 20th, celebrated annually since 2018 under the promotion of Slovenia and Apimondia [10], [11]. Both Apimondia and the Food and Agriculture Organization (FAO) actively encourage the harmonization of international honey quality criteria to combat trade fraud and ensure consumer safety [11], [12].

In Vietnam, while professional structures exist, such as the Beekeeping Association of Vietnam (established in 1981) and the Tropical Beekeeping Research and Development Center under the Vietnam National University of Agriculture (established in 2006), smallholder beekeeping practices remain largely unmechanized [13], [14]. To address this gap, Decision No. 898/QĐ-BNN-CN, signed on April 2, 2024, by the Ministry of Agriculture and Rural Development, laid out a sustainable development master plan for the beekeeping industry through 2030, emphasizing value chain digitalization, scientific-technological adoption, and export-compliant traceability [15], [16]. While IoT-based sensors have been tested experimentally in domestic academic beekeeping, commercial-scale digitization and traceability infrastructure remain highly limited [14], [17].

1.1.3. Market Solutions and Technological Entry Barriers

To monitor apiaries, several international IoT-based solutions have emerged, but they present significant financial and functional entry barriers for Vietnamese beekeepers [19]–[22]:

- BroodMinder (USA, est. 2014): Retails its entry-level "Apiary Starter Pack" at $549.99 USD (~13.7 million VND), accompanied by recurring cloud subscription fees of $50–$200 USD per year [19]. While structurally durable with a 5-year battery life, the high initial investment, English-only interfaces, and lack of local support make it less accessible to local smallholders.
- BeeHero (Israel/USA, est. 2017): This system focuses on large-scale industrial contract pollination, managing over 300,000 hives. However, it requires a complex local gateway infrastructure and high operational costs, which may not align with Vietnam's dispersed, nomadic beekeeping practices [21].
- BeeSage (Europe, est. 2015): Retails its 'HiveScale' system between $300 and $500 USD (~7.5–12.5 million VND) [22]. It is primarily focused on weighing scales and lacks localized distribution networks, making it difficult for Vietnamese farmers to access or maintain.

Thus, there exists a critical market gap for a localized, low-cost monitoring solution capable of non-invasive diagnostics. Non-invasive monitoring is vital because traditional physical hive inspections can alter internal hive conditions and stress the colony [23]. By tracking microclimatic deviations, such as fluctuations from the optimal 30–32°C temperature and 50–60% humidity thresholds, beekeepers can identify abnormal hive conditions at an early stage [23]. Furthermore, automated acoustic and vibration analysis can detect swarming behavior—characterized by pre-swarm weight increases and distinct sound patterns—allowing beekeepers to respond earlier and maintain stable colony conditions [24].

1.2 Need Statement

Object	Need Statement	Insight	Desired Outcome
Small-scale beekeeper	Needs a way to know which hives require attention without opening them	Manual hive inspection disturbs and stresses the colony, while abnormal conditions (queen loss, swarming, disease) are only discovered after it's too late	Reduce the frequency of physical inspections while catching problems early, without harming the colony
Export enterprise / cooperative	Needs real-time traceability data (GPS location, per-hive microclimate conditions) to standardize export documentation	Shipments get rejected due to missing provenance records (EU TRACES mandate, effective Nov 29, 2024), not due to actual product defects, and current lab-testing costs are high	Reduce verification/testing costs and increase the acceptance rate of export shipments while meeting HACCP/EU compliance

Consumer / honey buyer	Needs a way to verify the origin and trustworthiness of the honey being purchased	Consumers cannot distinguish fake/sugar-adulterated honey (63.6% concerned) or honey of unclear origin (48.5%) using appearance alone	Increase purchasing confidence and reduce the risk of buying low-quality honey
Agricultural officer / expert / policymaker	Needs a low-cost, easily scalable system to digitize the beekeeping sector in line with national policy	Existing international IoT solutions (BroodMinder, BeeHero, BeeSage) are expensive (7.5–13.7 million VND) and poorly suited to Vietnam's dispersed, nomadic beekeeping practices	Advance sector modernization under Decision 898/QĐ-BNN-CN (sustainable development plan through 2030)

1.3 Objectives statement

	Objective	Description
1	Non-invasive monitoring	Design a hive-monitoring system that eliminates the need for physical hive opening for routine condition checks, using external sensors (temperature, humidity, weight, sound, GPS) only.
2	Early risk detection	Enable the system to detect and alert beekeepers to abnormal hive conditions (queen loss, swarming, disease-related microclimate deviation, predator attack, theft) within hours rather than weeks.
3	Affordability for smallholders	Keep total hardware cost per hive within the range Vietnamese beekeepers are willing to pay (under ~1,000,000 VND/hive), well below existing international solutions (7.5–13.7 million VND).
4	Export-ready traceability	Provide time-stamped, GPS-linked hive records that can support documentation requirements for international honey export markets (e.g. EU TRACES-style traceability).
5	Field-viable deployment	Achieve sufficient battery life and reliable wireless connectivity to support real-world use in dispersed, nomadic beekeeping conditions typical of Vietnam.
6	Accessible reporting	Deliver hive data through a simple web/mobile dashboard usable by beekeepers with limited technical background, without requiring English-language interfaces.

Category	Key Functions	How to do	Objective
MONITOR	Temperature & humidity	Sensors measure the internal and external hive environment continuously.	Monitor hive microclimate and detect abnormal conditions.
	Hive weight	A load-cell system measures changes in hive weight to track honey accumulation and colony mass.	Monitor colony development and honey production.
	Hive location	GPS records the geographic position of the hive.	Track hive location and support traceability.
	Bee traffic	Sensors detect the activity/rate of bees entering and leaving the hive.	Observe colony activity and behavioral changes.
	Physical disturbance	Motion/impact sensing detects unusual movement, falls, or possible external disturbance.	Provide early warning of physical risks.
	Battery level	The system monitors battery voltage/power status.	Prevent unexpected system shutdown.
CONTROL	Sensor sampling	The MCU controls when and how frequently each sensor collects data.	Ensure coordinated and efficient data acquisition.

	Audio gain	The audio circuit adjusts the signal level before acoustic data is processed.	Obtain usable hive sound for AI analysis.
	MCU processing	The microcontroller manages sensor reading, data processing, communication, and AI tasks.	Coordinate system operations efficiently.
	Wireless/FOTA updates	Firmware can be updated remotely through wireless communication.	Enable remote maintenance without physical access.
OPTIMIZE	AI model size	The AI model is optimized and deployed on the edge device with limited memory.	Reduce memory and hardware requirements.
	Inference speed	AI inference is optimized to process hive data locally with low latency.	Enable fast abnormal-condition detection.
	Data transmission	Only useful sensor data and AI results are transmitted instead of continuously sending raw data.	Reduce bandwidth and communication overhead.
	Power consumption	Sampling, processing, and communication are managed to reduce unnecessary energy usage.	Extend battery operating time.
	Weight accuracy	The weighing platform is designed to maintain balanced loading and reliable measurements.	Improve accuracy of hive-weight monitoring.

AUTOMATE	Queen-state classification	AI analyzes hive acoustic/sensor patterns to classify the colony state related to queen presence.	Reduce the need for frequent manual inspection.
	Risk detection	Sensor and AI data are analyzed to identify abnormal conditions and potential risks such as swarming or colony problems.	Provide early warnings to beekeepers.
	Cloud synchronization	Sensor measurements and AI results are automatically uploaded to the database/cloud.	Maintain centralized, up-to-date hive records.
	Traceability reports	Location, sensor history, and hive information are linked into records for monitoring and reporting.	Support long-term management and traceability.

Reference

[1] Nhân Dân, “Đưa công nghệ số phát triển ngành ong Việt,” 2025. [Online]. Available: https://nhandan.vn/dua-cong-nghe-so-phat-trien-nganh-ong-viet-post884472.html

[2] Bộ Nông nghiệp và Phát triển Nông thôn, “Báo cáo thống kê ngành ong,” 2024. [Online]. Available: https://www.mard.gov.vn

[3] Báo Gia Lai, “Biến đổi khí hậu khiến loài ong mật đối mặt với nhiều mối đe dọa,” May 2024. [Online]. Available: https://baogialai.com.vn/bien-doi-khi-hau-khien-loai-ong-mat-doi-mat-voi-nhieu-moi-de-doa-post324181.html

[4] Nông Nghiệp Môi Trường, “Thuốc trừ sâu khiến ong mật vài thế hệ để phục hồi,” Jul. 2024. [Online]. Available: https://nongnghiepmoitruong.vn/thuoc-tru-sau-khien-ong-mat-vai-the-he-de-phuc-hoi-d308674.html

[5] Dân Việt, “Xuất khẩu mật ong 9 tháng đạt 24.000 tấn, Mỹ thu mua nhiều nhất,” Oct. 2022. [Online]. Available: https://danviet.vn

[6] European Commission, “Technical Guidelines for Determining the Magnitude of Pesticide Residues in Honey (SANTE/11956/2016 rev.9),” 2016. [Online]. Available: https://food.ec.europa.eu/document/download/06bcc245-57b5-47ff-b31b-25c05a8485cf_en?filename=pesticides_mrl_guidelines_honey.pdf

[7] European Commission, “Regulation (EU) 2023/2652 on Registration of Bee Products in TRACES,” Nov. 2023.

[8] CBI – Centre for the Promotion of Imports, “What Requirements Must Honey Comply with to Be Allowed on the European Market?” 2023. [Online]. Available: https://www.cbi.eu/market-information/honey/what-requirements-should-your-product-comply

[9] USDA Agricultural Marketing Service, “Honey Import Requirements and HACCP Compliance,” 2024. [Online]. Available: https://www.ams.usda.gov/sites/default/files/media/AA20380_Honey.pdf

[10] United Nations, “World Bee Day – 20 May.” [Online]. Available: https://www.un.org/en/observances/bee-day

[11] Food and Agriculture Organization of the United Nations, “World Bee Day.” [Online]. Available: https://www.fao.org/world-bee-day/en

[12] Apimondia, “Apimondia Statement on Honey Fraud.” [Online]. Available: https://www.apimondia.org/uploads/7/6/4/3/76436993/apimondia_statement_on_honey_fraud.pdf

[13] Hội Nuôi ong Việt Nam, “Giới thiệu về Hội Nuôi ong Việt Nam,” 1981. [Online]. Available: https://hoiong.org.vn

[14] Trung tâm Nghiên cứu Ong và Nuôi ong Nhiệt đới, Học viện Nông nghiệp Việt Nam, "Ứng dụng công nghệ 4.0 trong nuôi ong," 2006. [Online]. Available: https://hvnvn.vn

[15] Bộ Nông nghiệp và Phát triển Nông thôn, "Quyết định số 898/QĐ-BNN-CN ngày 02/4/2024," 2024. [Online]. Available: https://thuvienphapluat.vn/van-ban/Linh-vuc-khac/Quyet-dinh-898-QD-BNN-CN-2024-phe-duyet-De-an-Phat-trien-ben-vung-nganh-Ong-den-2030-625836.aspx

[16] Bộ Nông nghiệp và Phát triển Nông thôn, "Đề án Phát triển bền vững ngành Ong đến 2030," 2024.

[17] Nông nghiệp & Môi trường, "Ngành ong mật chuyển dịch bằng số hóa," 2024. [Online]. Available: https://nongnghiepmoitruong.vn/nganh-ong-mat-chuyen-dich-bang-so-hoa-d383804.html

[18] Nông nghiệp & Môi trường, "Ngành ong mật chuyển dịch bằng số hóa," 2024.

[19] BroodMinder, "BroodMinder-BeeDar Sensor Products." [Online]. Available: https://broodminder.com/collections/sensors/products/broodminder-beedar

[20] BroodMinder, "BroodMinder-W5 Weight Scale." [Online]. Available: https://broodminder.com/collections/sensors/products/broodminder-w5-weight-scale

[21] BeeHero, "Beekeeping Solutions – IoT and AI for Beekeeping." [Online]. Available: https://www.beehero.io/beekeeping-solutions

[22] BeeSage, "HiveScale – Smart Hive Monitoring System." [Online]. Available: https://beesage.co/en/hivescale

[23] H. Zhang *et al*., "Smart beehive monitoring system based on IoT sensors," *Computers and Electronics in Agriculture*, vol. 168, 2020, pp. 1–12.

[24] M. Ramsey *et al*., "Swarming detection in honeybee colonies using vibration analysis," *IEEE Transactions on Industrial Electronics*, vol. 67, no. 4, pp. 3421–3431, 2020.